MINI PROJECT 1

BUILD YOUR OWN CYBERSECURITY LAB

Beginner-friendly virtual lab guide

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Educational mini project

Project Overview

This mini project explains how to create a beginner-friendly cybersecurity lab using Oracle VirtualBox and Kali Linux. A virtual machine lets you practice in an isolated environment without replacing the main operating system on your computer. The screenshots in this document are taken from the supplied “balaji mini project.pdf” and placed alongside the matching workflow where possible.

Before You Begin

• Check that your computer has enough available memory and disk space.

• Download software only from the official publisher websites.

• Keep your host operating system updated and back up important files.

• Use a strong password for the virtual machine.

• Only test cybersecurity tools on systems and networks you own or have permission to assess.

Official Downloads

Oracle VirtualBox: https://www.virtualbox.org/wiki/Downloads

Kali Linux: https://www.kali.org/get-kali/

Windows 11 (optional separate VM): Microsoft official Windows 11 download

Part 1 — Install Oracle VirtualBox

STEP 1 Open the official VirtualBox download page

Open a browser, visit the official VirtualBox downloads page, and choose the package for your host operating system.

Miniproject-1

Mini project

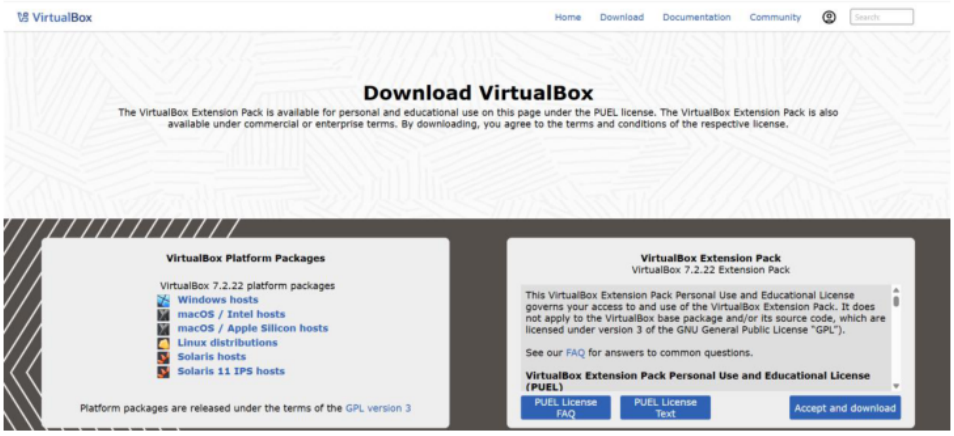
Module -1: setup & Build Your Own Cybersecurity Lab

Step1:

Open the chrome search for virtual box in chrome

Download virtual box

https://www.virtualbox.org/wiki/Downloads



Select the which devices to download of the virtual box

And accept and download the extension of virtual box

When we select that option download automatically

Step 2:

Search for the kali in chrome

https://www.kali.org/get-kali/#kali-platforms

download the kali

Original source page 1 — VirtualBox download instructions

STEP 2 Choose the appropriate download

Select the correct VirtualBox download for your computer. The original project also describes the Extension Pack; it is optional and should be installed only if you need its extra features.

Miniproject-1

Mini project

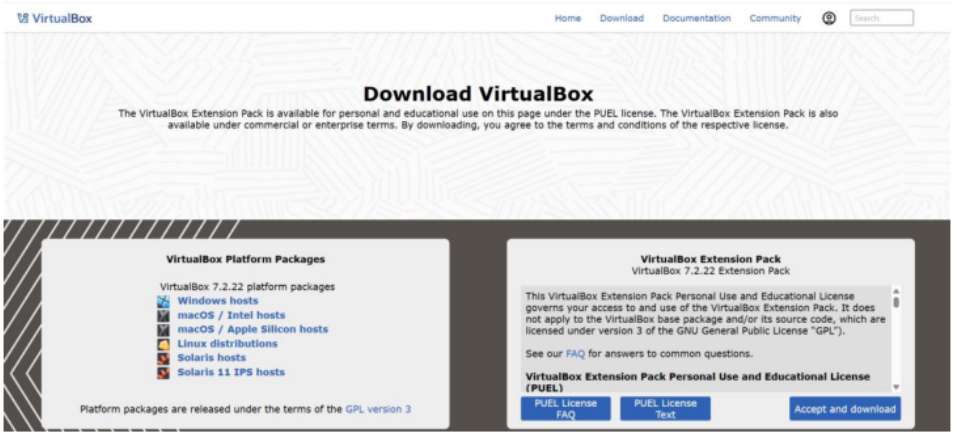
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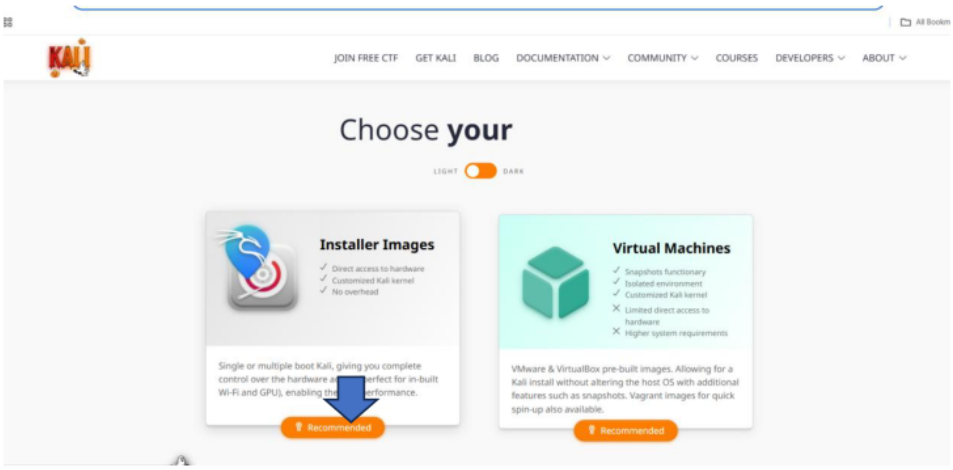
download the kali

Original source page 1 — Download and Extension Pack notes

Part 2 — Download Kali Linux

STEP 3 Open the official Kali Linux website

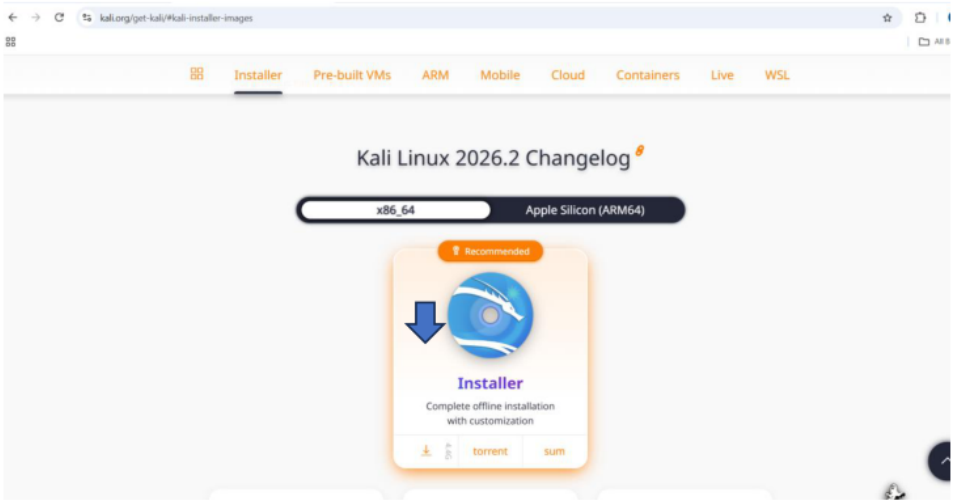
Visit the official Kali Linux download page. Choose the recommended installer image suitable for your computer, and save the ISO file somewhere easy to locate.



Recommender option

It another option website

https://www.kali.org/get-kali/#kali-installer-images



Click that option

When Download completes it shown download

Original source page 2 — Kali Linux download options

STEP 4 Wait for the download to complete

Confirm that the ISO download has finished before creating the virtual machine. The ISO is the installation image used by VirtualBox.



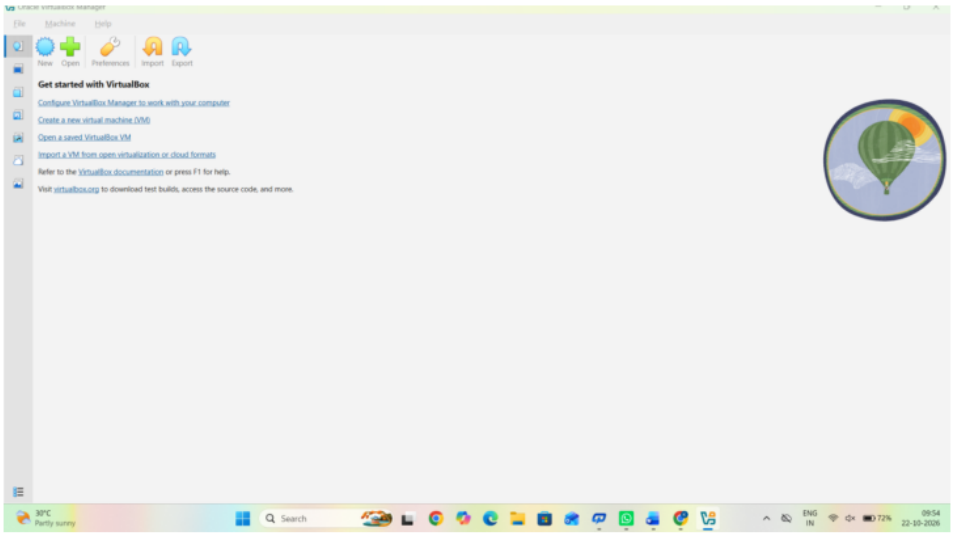
After completing it show like this only

Step 3:

Open file manger

Search for virtual box and open virtual box

Click on the accept next button until it finish



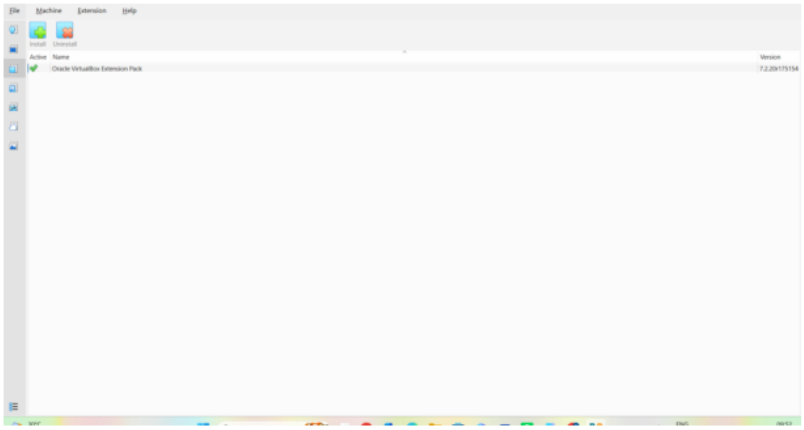
Click on extension button and add extension through file manger

Original source page 3 — Download completion / file view

Part 3 — Create and Configure the Kali VM

STEP 5 Open VirtualBox and complete setup

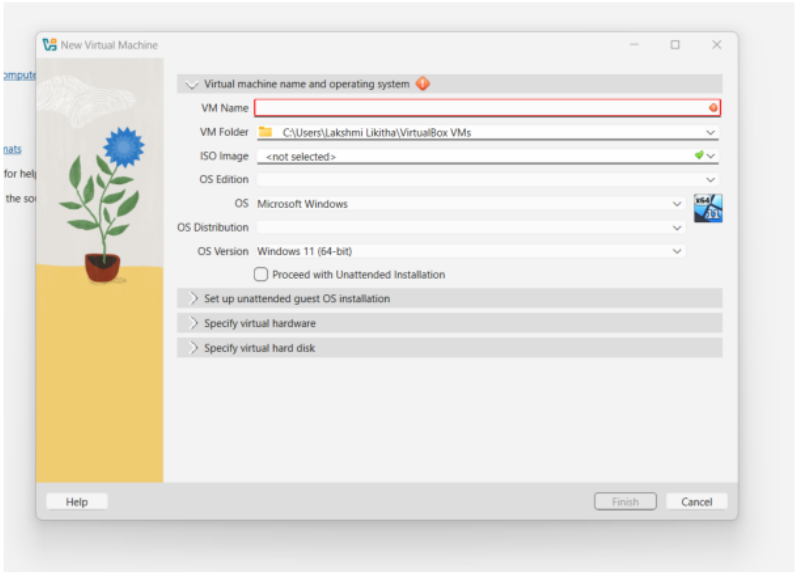
Open the file manager, launch VirtualBox, and follow the installer prompts. If you downloaded the Extension Pack, open VirtualBox's extension settings and add the downloaded file.



After completing it shows above image

After that click on home button

And new button



Open and setup the file name

Iso image

Os edition

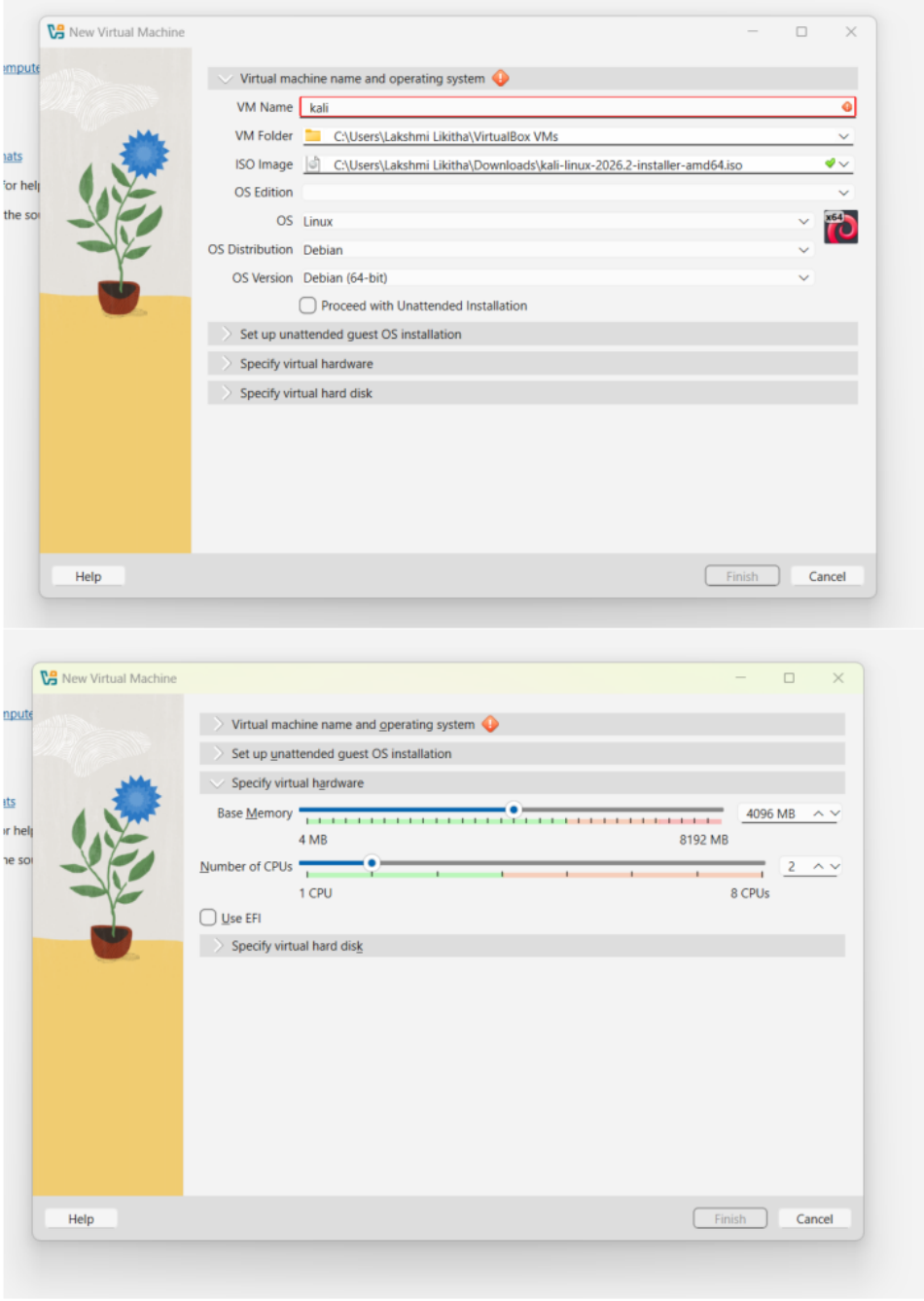
Os distribution

Os version of it show in the below image

Original source page 4 — VirtualBox setup

STEP 6 Create a new virtual machine

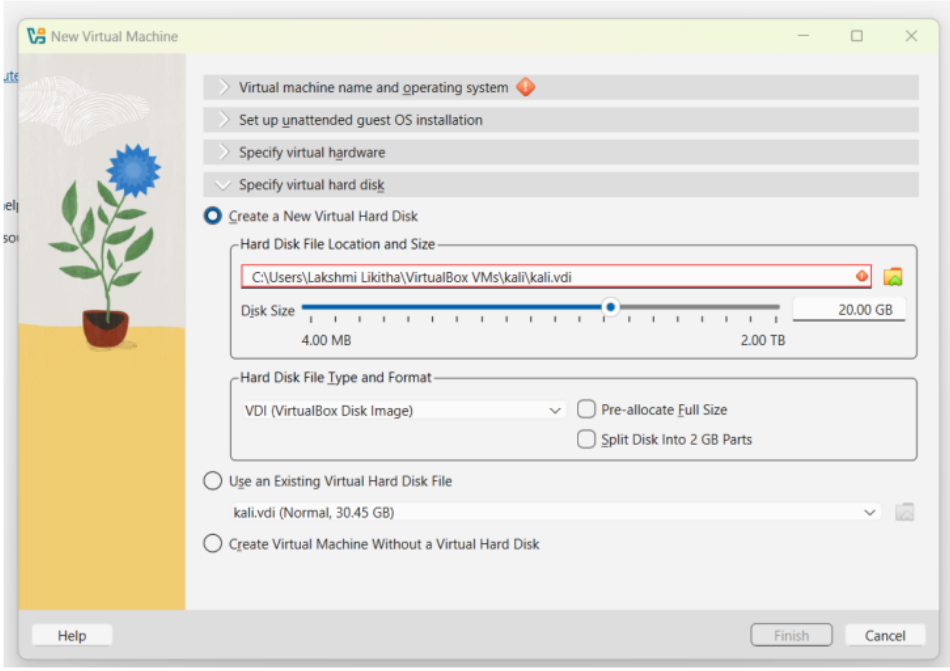
Select Home, then New. Enter a name for the VM and select the Kali ISO image. Check the operating system type and version shown by VirtualBox.



Original source page 5 — New VM configuration

STEP 7 Finish the configuration

Review the VM settings and click Finish when the button becomes available. Allocate memory, CPU, and virtual disk space sensibly for your computer; keep enough resources for the host operating system.



After the finish button enable click on it

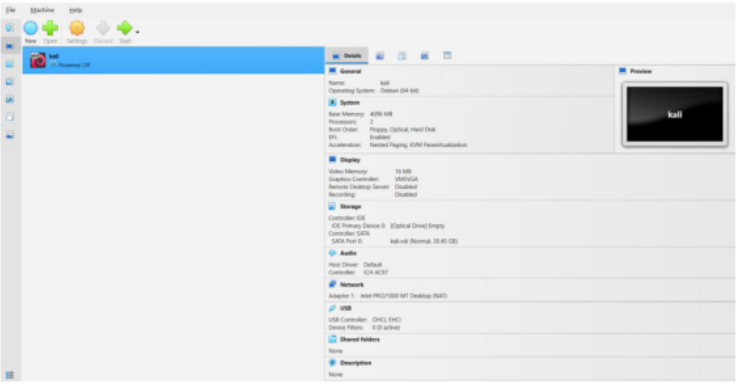
Remaining after that

User:xxxxx

Password :xxxxx

Click on the yes button

Until it finish setup



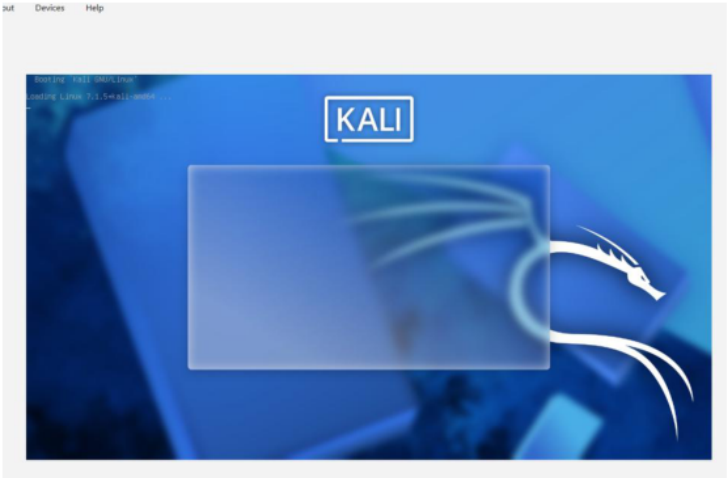
Click on the power on button

Original source page 6 — Finish and initial configuration

Part 4 — Start Kali and Verify the Installation

STEP 8 Power on the virtual machine

Select the Kali VM and click Start. Allow the system a few minutes to boot, then sign in using the username and password you created during installation.



Wait until 5 -minutes

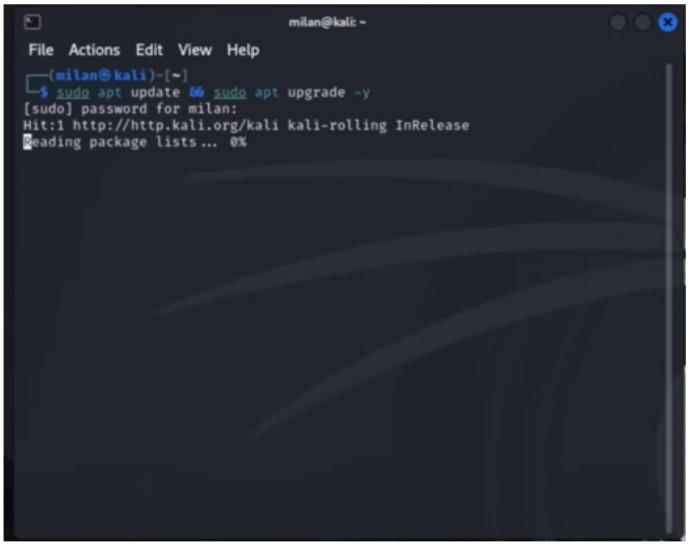
Enter user name :

Password:

Open the terminal of kali

Run this command on it

Sudo apt update && sudo apt upgrade command on terminal

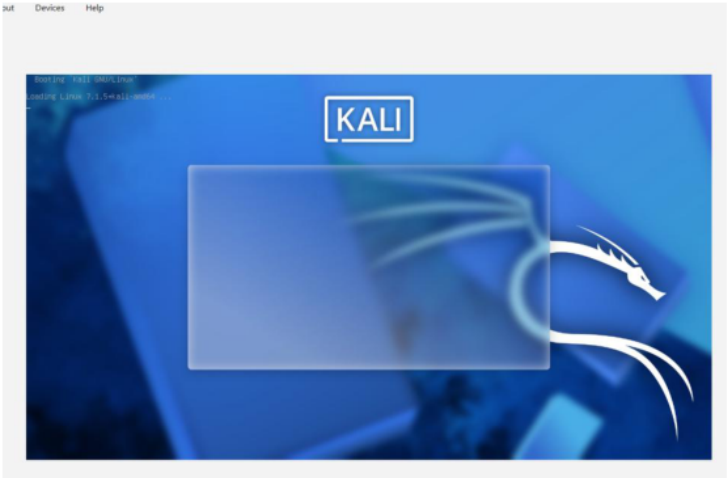


After this we can run another command:cat /etc/os-release

Original source page 7 — Starting the virtual machine

STEP 9 Update the Kali package list

Open the Kali terminal and run the command below. Review any package changes before confirming an upgrade.



Wait until 5 -minutes

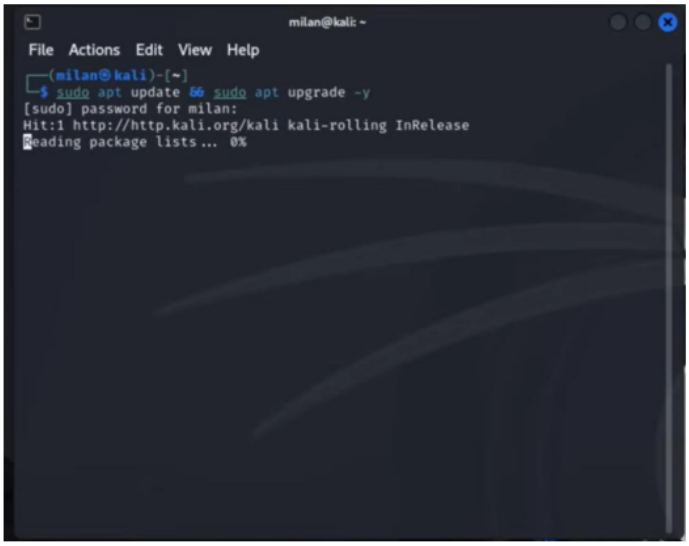
Enter user name :

Password:

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Run this command on it

Sudo apt update && sudo apt upgrade command on terminal



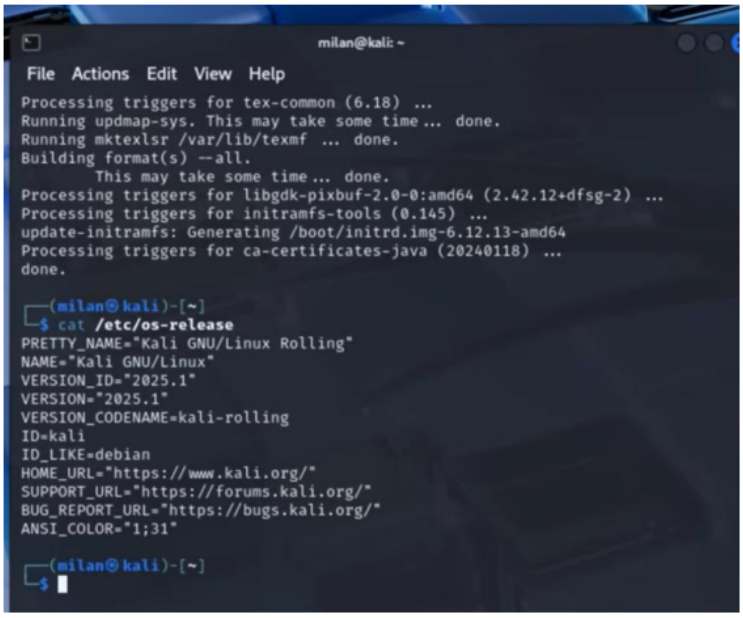
After this we can run another command:cat /etc/os-release

Original source page 7 — Terminal and update instructions

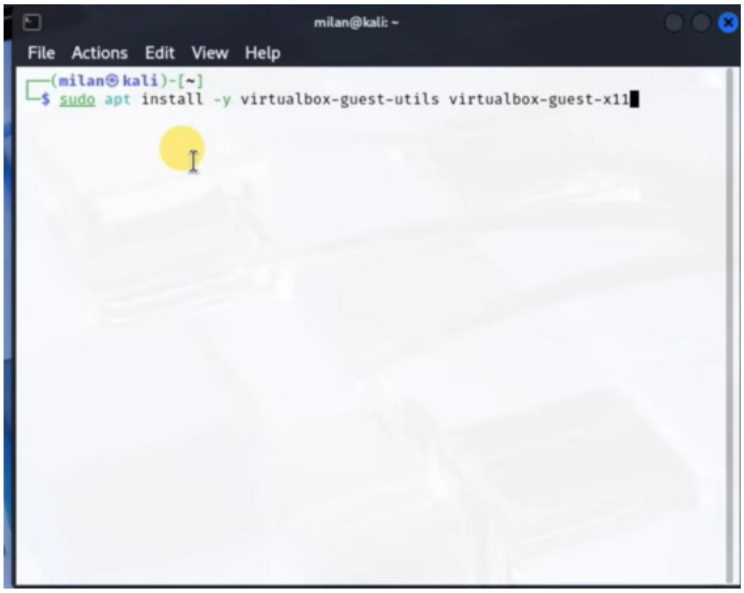
```
sudo apt update && sudo apt upgrade
```

STEP 10 Check the operating system information

To view system release details, run this command in the Kali terminal:



It shown as version of the windows



Original source page 8 — System information

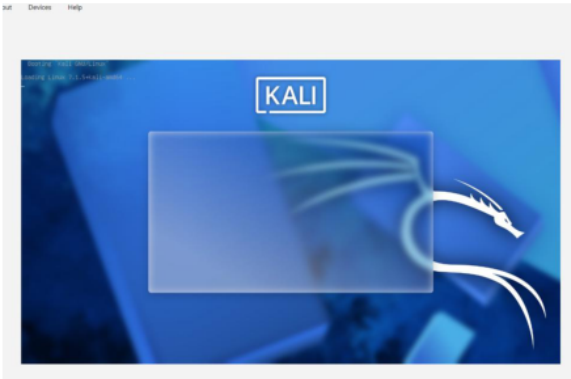
```
cat /etc/os-release
```

STEP 11 Confirm the desktop is working

After the system starts successfully, check that the Kali desktop and terminal are accessible. If the VM returns to the installer, check that the ISO is detached or that the virtual hard disk is selected first in the boot order.

After this command we should our system

And the interface will be



After that enter user name :

Password ;

Of it and final our Build Your Own Cybersecurity Lab will be done here

Successfully completed the own cybersecurity lab it will be done here

Conclusion:

Hence, completely finally Build Your Own Cybersecurity Lab

Original source page 9 — Final lab interface

Optional — Create a Separate Windows VM

If your project requires both Kali Linux and Windows, create a second virtual machine rather than installing Windows inside the Kali VM. Download the Windows ISO only from Microsoft's official website. Follow the Windows setup prompts and check applicable licensing requirements.

Official Windows 11 download: https://www.microsoft.com/en-in/software-download/windows11

Keep the lab safe

• Use snapshots before experiments so you can restore a known-good state.

• Keep the host and guest systems updated.

• Use isolated or host-only networking for practice when appropriate.

• Never scan, access, or attack networks or devices without explicit permission.

Conclusion

Hence, the cybersecurity lab setup is completed by preparing VirtualBox, downloading the Kali Linux ISO, creating a virtual machine, installing Kali, and verifying the desktop and terminal. This environment provides a place to learn cybersecurity concepts responsibly.

Screenshot source: user-provided "balaji mini project.pdf". Some source pages contain multiple elements; screenshots are reproduced as page images for reference.